Welcome aboard

Modern Windows Security Bypassed

Sergey Chubarov & Morten Knudsen

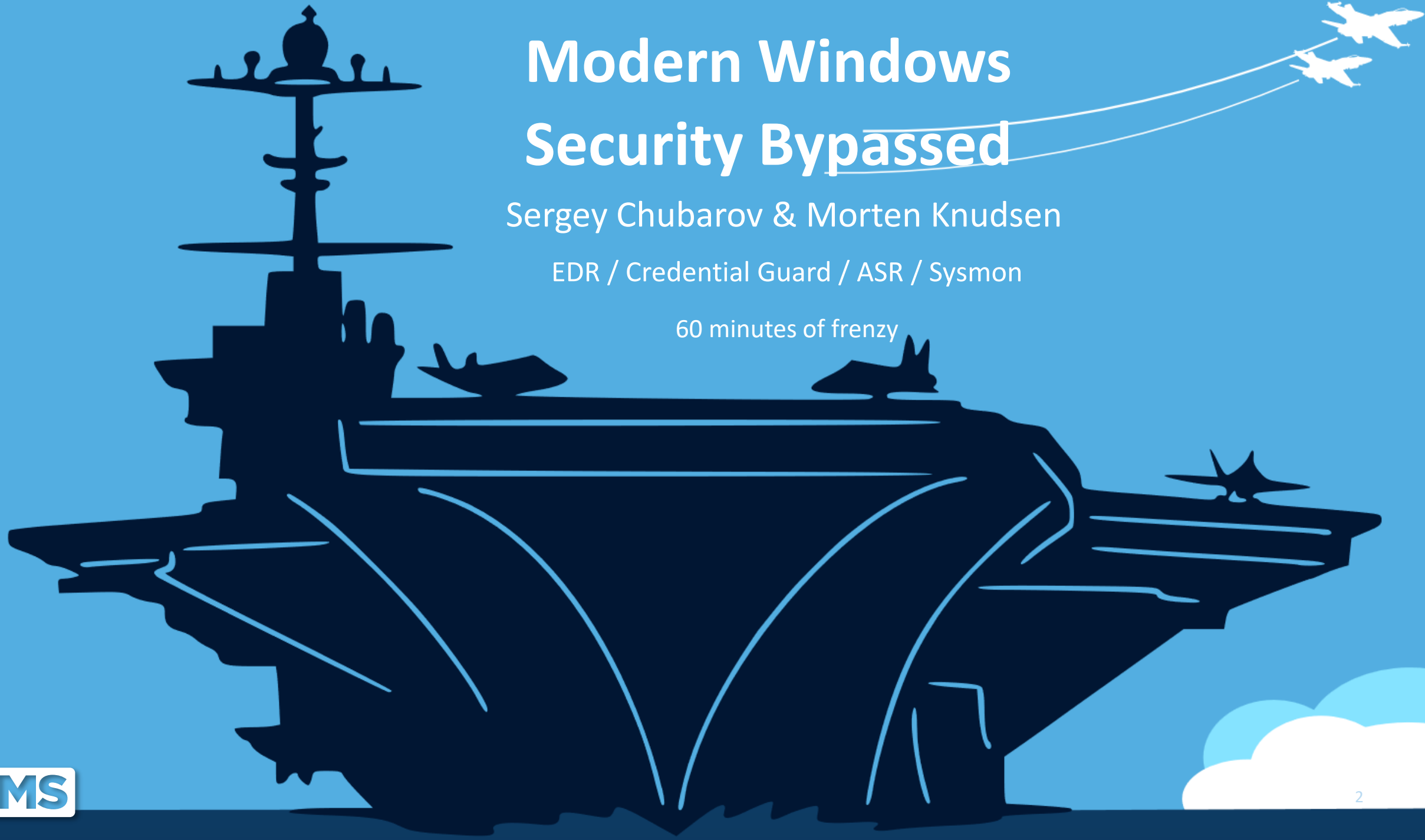


Modern Windows Security Bypassed

Sergey Chubarov & Morten Knudsen

EDR / Credential Guard / ASR / Sysmon

60 minutes of frenzy



Thank you to our sponsors!



Speakers



Sergey Chubarov

MVP Security | Russian Hacker



Morten Knudsen

Triple MVP
Security, Azure, Security Copilot

Since 2021 we've shown you

And no admin rights required

Defender Antivirus

BYPASSED

Attack Surface Reduction (via alternative path)

BYPASSED

Controlled Folder Access

BYPASSED

Defender for Office 365

BYPASSED

AppLocker

BYPASSED

AMSI in Windows 10 & 11

BYPASSED



Local admin is no longer the master switch

TAMPER PROTECTION

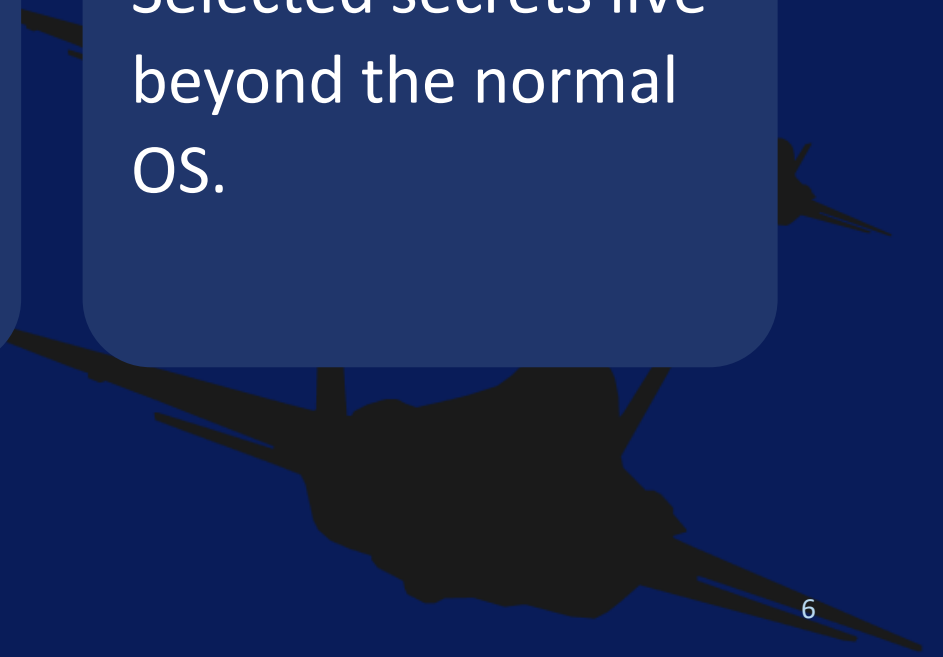
Security settings resist ordinary local changes.

PPL

Protected processes restrict access and modification.

VBS

Selected secrets live beyond the normal OS.

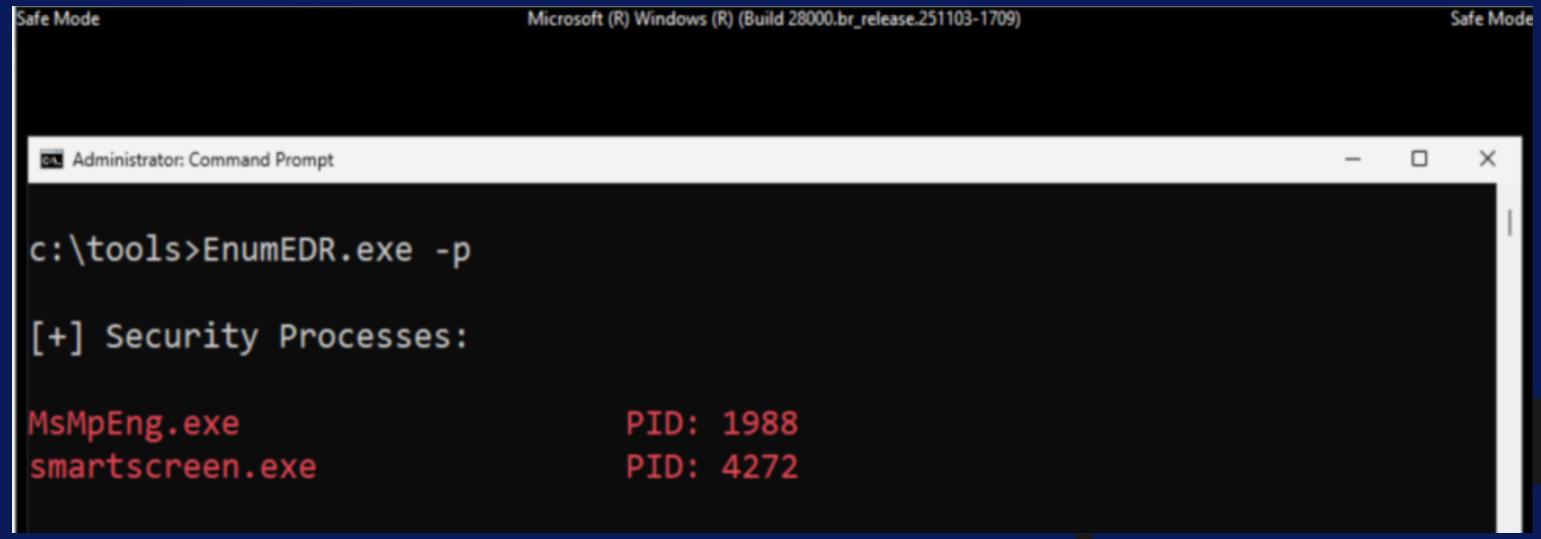


Stronger boundaries change the attacker's route.

Safe Mode ≠ defenses off

OLD ASSUMPTION

Minimal boot.
No security controls.



When the off-switch fails, change the route

DISCONNECT

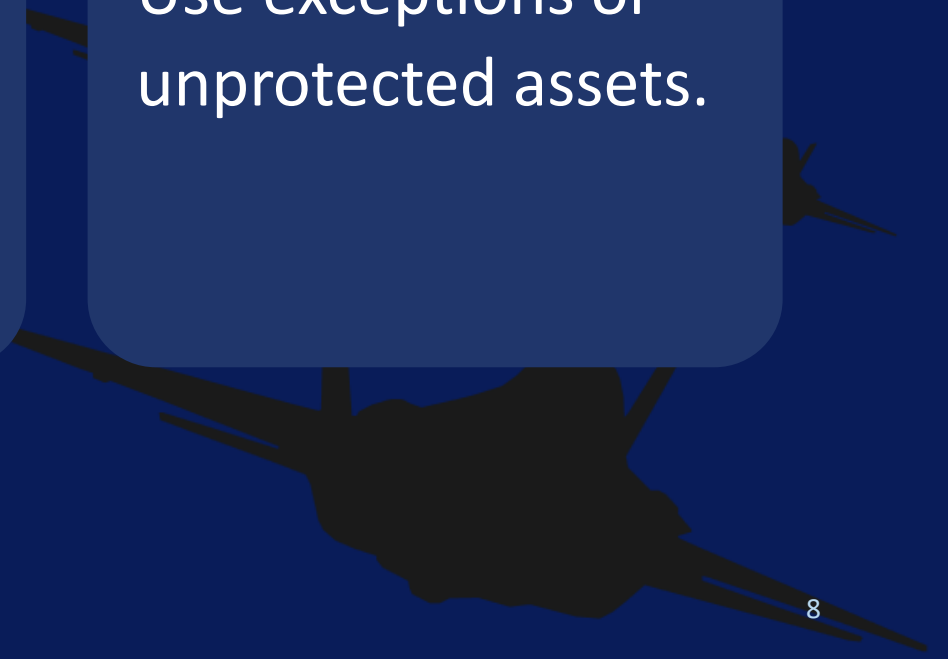
Break the reporting path.

IMPAIR

Abuse privileged components.

GO AROUND

Use exceptions or unprotected assets.



ASR Rule Bypass Techniques

03 / ATTACK SURFACE REDUCTION



ASR rules block specific behaviors

APPLICATION

Who performs the action?

›

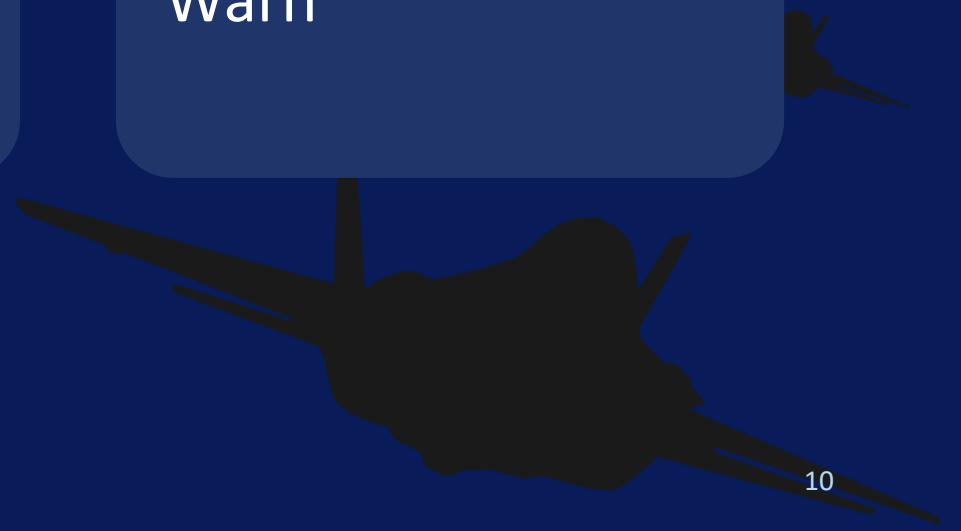
BEHAVIOR

What does the rule match?

›

DECISION

Block / Audit / Warn



A rule is not a universal execution policy.

Inside the rule: compiled Lua and exceptions

VDM / .LUAC

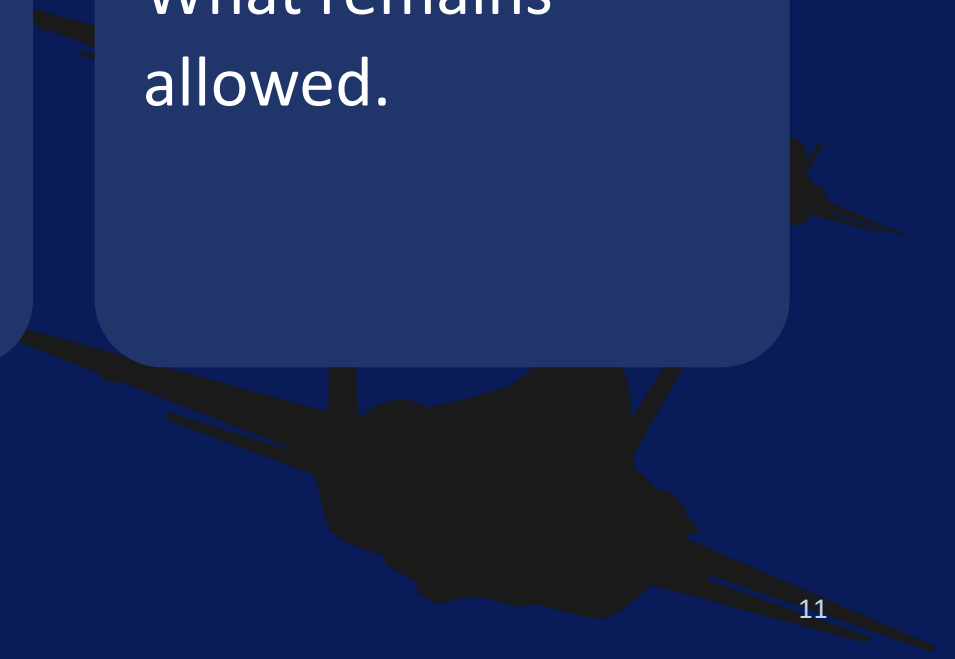
Compiled rule logic.

PREDICATES

What the rule evaluates.

EXCEPTIONS

What remains allowed.



Three meanings of "excluded"

BUILT-IN

Compatibility logic inside the product.

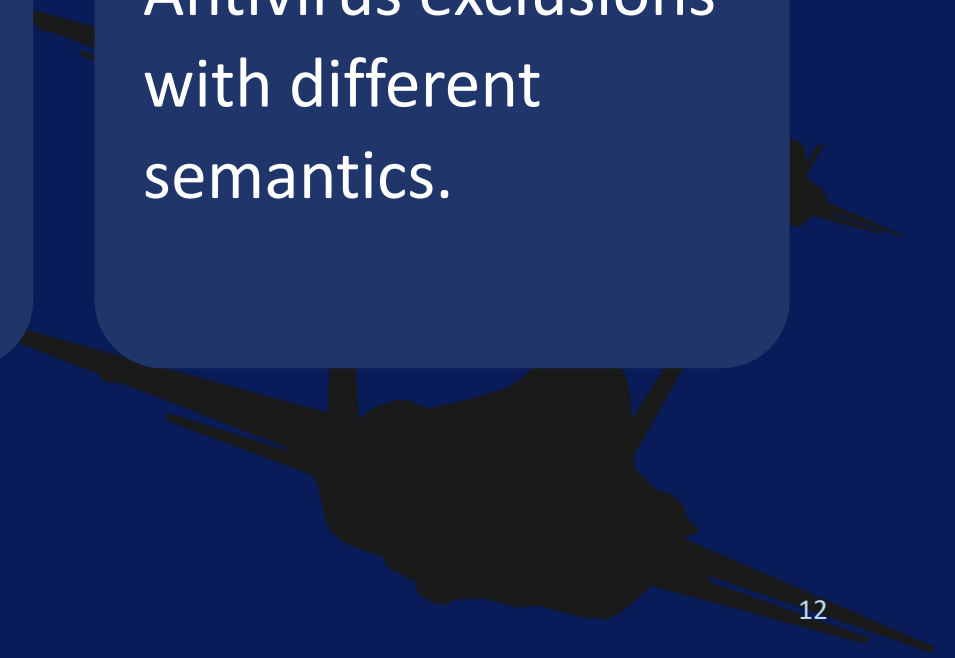
ASR POLICY

Exceptions configured by the organization.

AV POLICY

Antivirus exclusions with different semantics.

Different exceptions. Different mitigations.



Close the path the rule leaves open

NARROW EXCEPTIONS

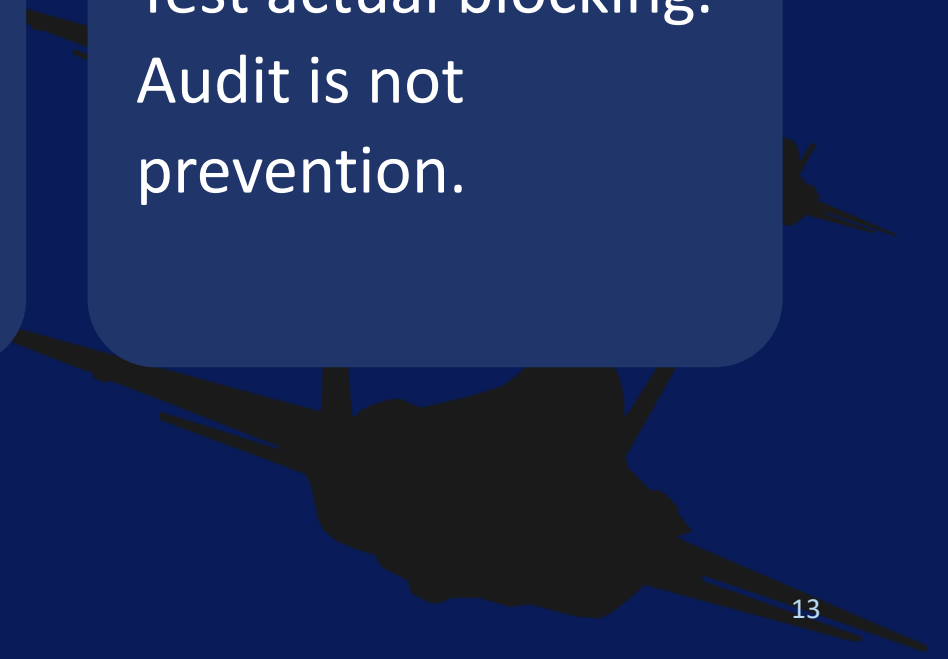
Scope. Owner. Expiry.

CONSTRAIN EXECUTION

App Control. Only required capabilities.

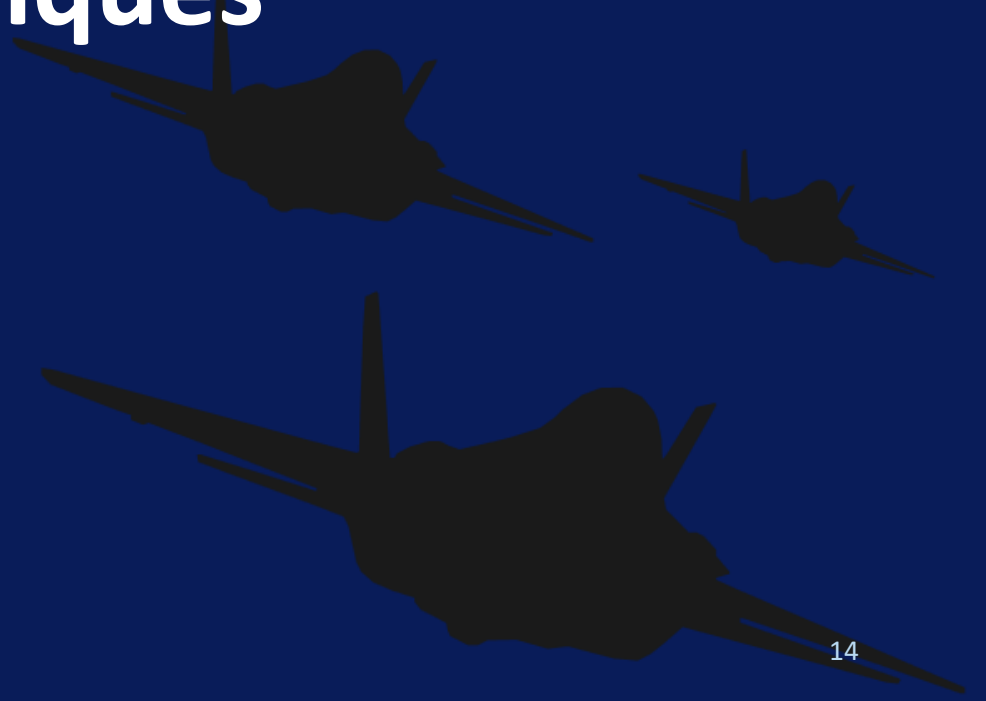
VERIFY THE ACTION

Test actual blocking. Audit is not prevention.



EDR Logic and Evasion Techniques

02 / ENDPOINT DETECTION & RESPONSE



EDR is a chain—not one process

SENSOR

Endpoint activity

›

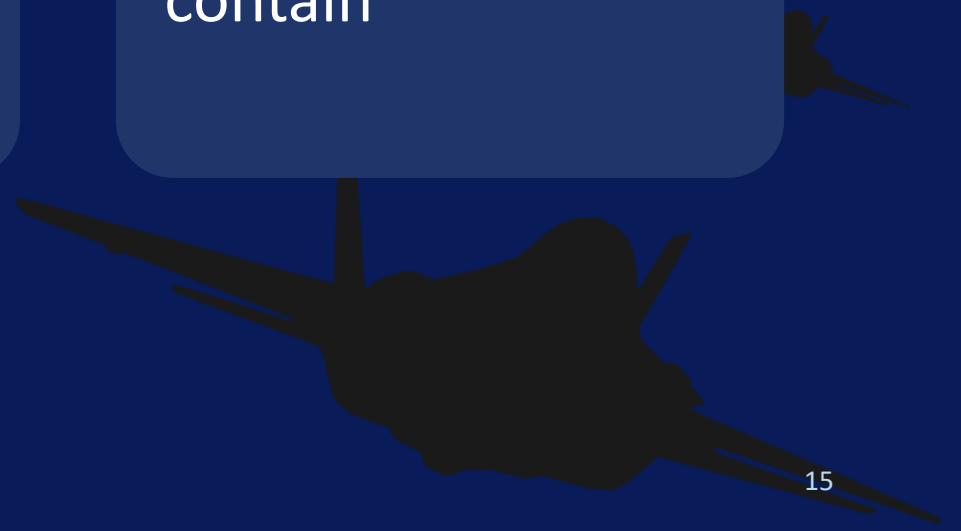
CLOUD

Behavior + context

›

RESPONSE

Investigate + contain



Firewall: disconnect the cloud

IMPACT

Telemetry delayed. Remote response impaired.

HARDENING

Protect network policy: disable local firewall rules. Alert on stale endpoints.

Cloud isolation ≠ local protection disabled.



Signed ≠ safe

PUBLISHER

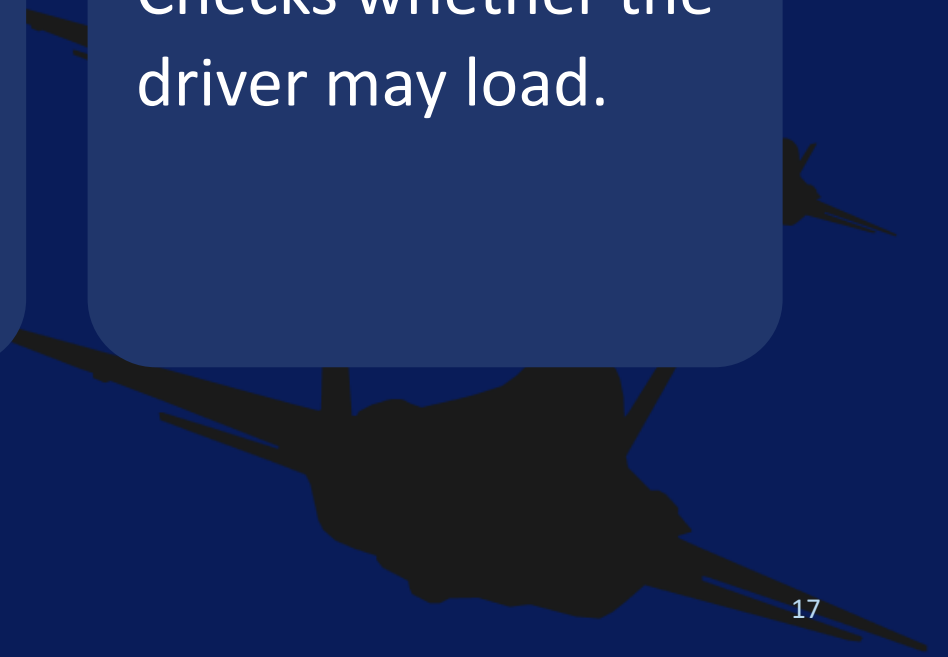
Develops the driver.

MICROSOFT

Signs new drivers through the hardware program.

CODE INTEGRITY

Checks whether the driver may load.



A trusted signature can belong to a vulnerable driver.

Terminator

02 / EDR • DEMO

EDR-killing tool

ZEMANA

zam64.sys / zamguard64.sys

Vulnerable anti-malware drivers



Kernel-assisted termination of security processes.

KslD.sys: Defender as the access path

DEFENDERDUMP / PUBLISHED RESEARCH

Local admin. Existing Microsoft driver. Sensitive memory.

KslD.sys

Microsoft-signed; Defender-supplied

Physical + virtual memory reads

No external driver in the reported scenario

- **Kernel addresses / KASLR**
- **PPL-protected process memory**
- **Process-token enumeration**

Used by KslKatz to dump credentials

Source: Pumi96 / DefenderDump

RTCore64: a kernel Swiss Army knife

BYOVD / MSI AFTERBURNER / CVE-2019-16098

A signed driver. Multiple paths to kernel abuse.

RTCore64.sys

Arbitrary memory read / write

I/O ports + CPU registers (MSRs)

- **Privilege escalation**
- **Security-control tampering**
- **Information disclosure**

More than process termination: a reusable kernel-access primitive.

Stop the driver before it stops the sensor

Driver Signature Enforcement (DSE)

Blocks unsigned or untrusted kernel drivers.

Attack Surface Reduction (ASR)

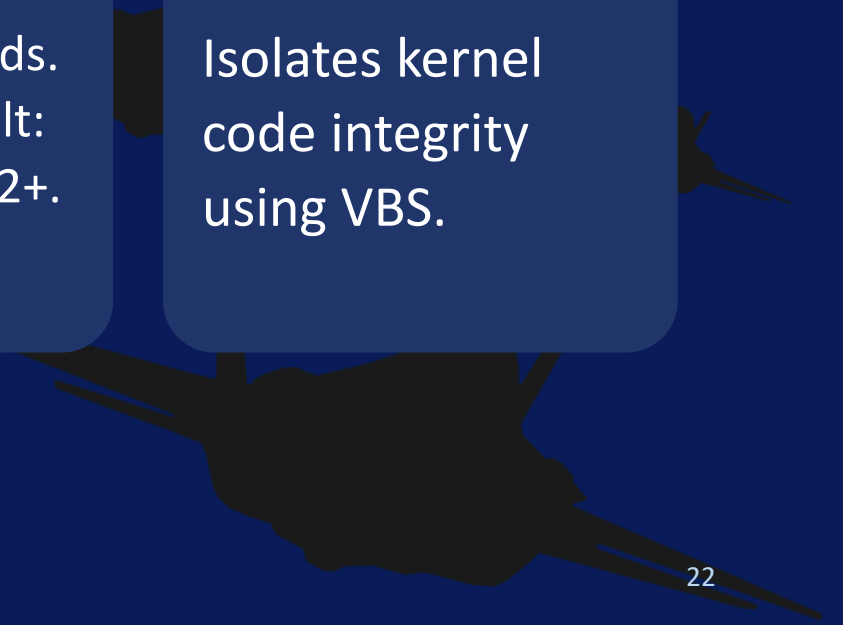
Blocks writes of known vulnerable signed drivers.

Microsoft vulnerable driver blocklist

Blocks unsafe loads. Enabled by default: Windows 11 22H2+.

Memory Integrity (HVCI)

Isolates kernel code integrity using VBS.



Write blocking and load blocking solve different steps.

Secure Boot and Memory Integrity

SECURE BOOT

Trust the startup chain.

BITLOCKER+TPM

Prevents automatic access to the encrypted drive after boot setting changes

Add driver blocking. Keep firmware and Windows current.

Testing Credential Guard Protections

04 / CREDENTIAL GUARD



Protect the secret—not every use of identity

NORMAL WINDOWS		VBS ISOLATION
Apps + authenticated sessions	↔	LSAIso
LSASS		**Protected NTLM hashes + TGTs**

Authentication can work without exposing the protected secret.

PPL ≠ Credential Guard

LSASS / PPL

Restricts process access.

CREDENTIAL GUARD

Isolates selected secrets.

Removing PPL does not automatically disable LSAIso.

Credential exposure outside LSAIso

NORMAL-OS PATH

LSASS protection. WDigest credential handling.

ISOLATED STORE

Credential Guard can remain a separate boundary.



Reduce exposure around the isolated store

PROTECT THE PLATFORM

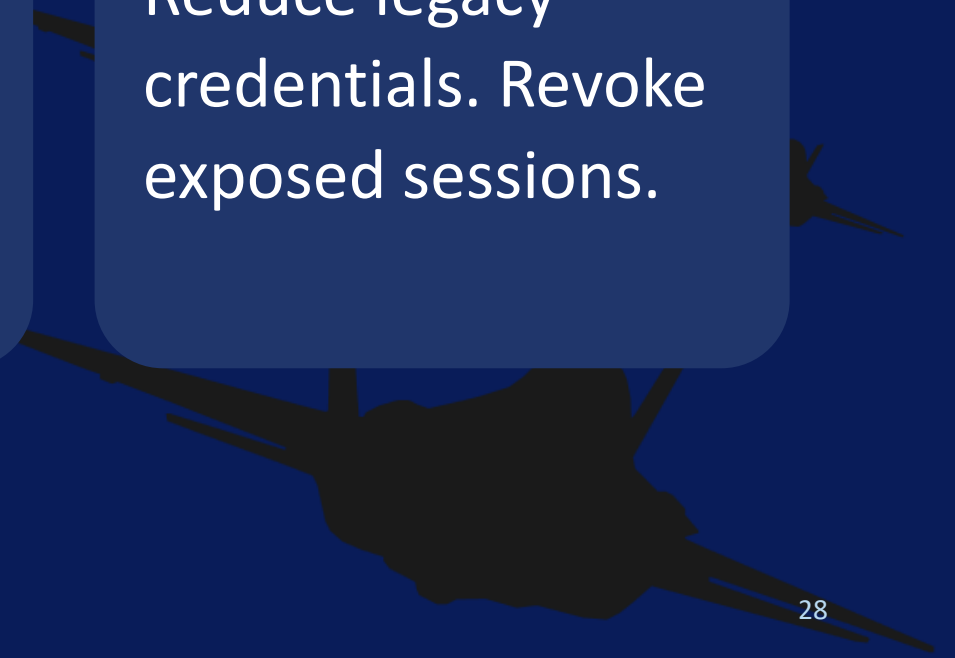
LSA protection. VBS. Restrict debug access.

SEPARATE ADMIN

Keep privileged sessions off everyday endpoints.

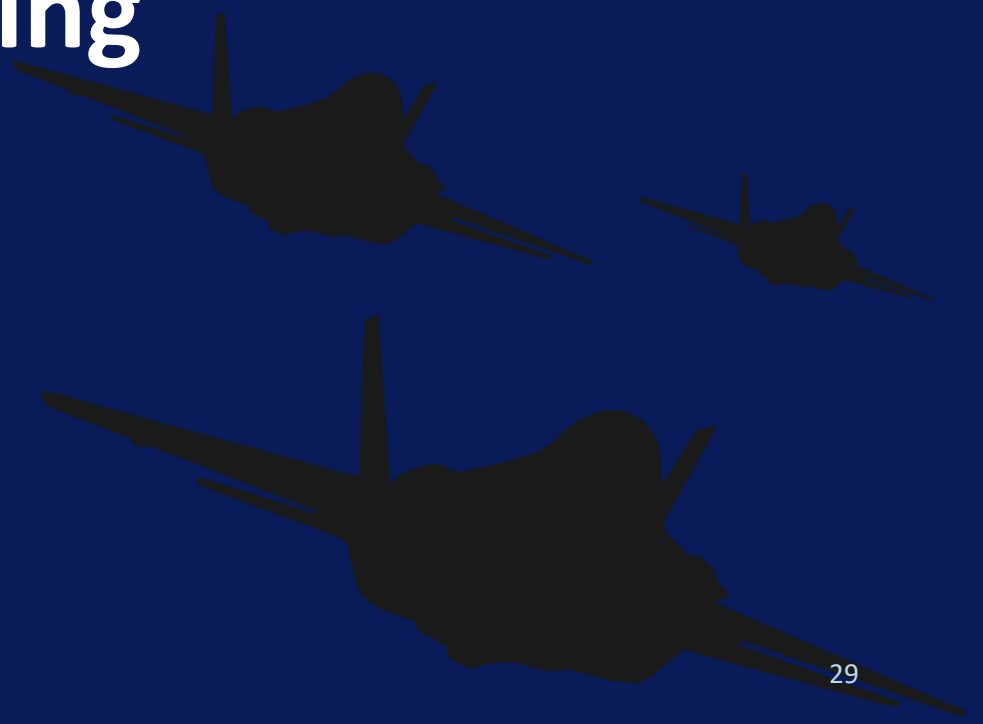
FIX THE AUTH PATH

Reduce legacy credentials. Revoke exposed sessions.



Sysmon Evasion and Tampering

05 / TELEMETRY & EVIDENCE



Sysmon records. Your pipeline detects.

CAPTURE

Service + driver

›

RECORD

Local event log

›

FORWARD

Collector / SIEM



Attack capture—or attack delivery

CAPTURE

Driver impairment.
Configuration gaps.

DELIVERY

Forwarding interrupted.
Events never reach the SOC.



A startup change can become a logging gap

BEFORE REBOOT

Startup config changes. Driver still loaded.

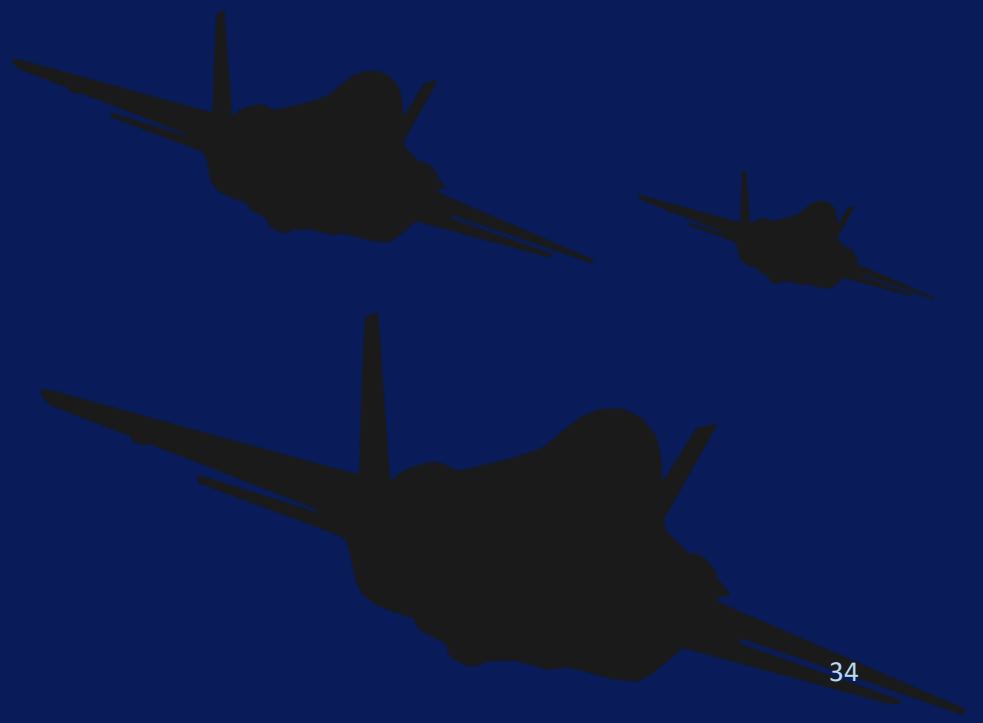
AFTER REBOOT

Driver may not start. Coverage can disappear.

Watch configuration changes-and the next boot.

Closing the gaps

06 / HARDENING & TAKEAWAYS



Remove the options attackers do not need

OLD DRIVERS

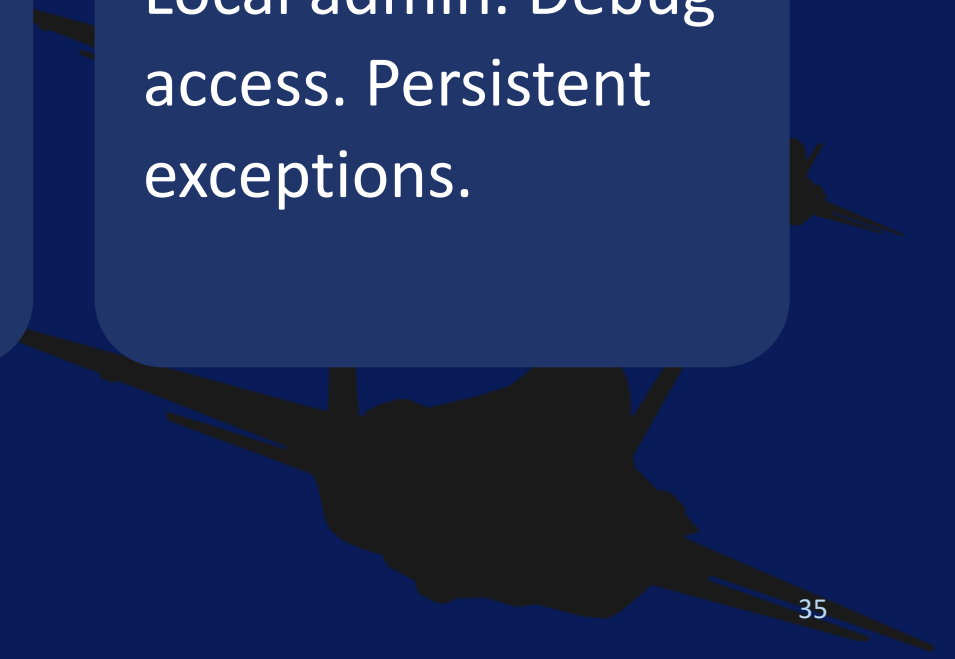
Retire unsafe versions and privileged utilities.

UNUSED FEATURES

Macros. Script hosts. Unnecessary remote access.

EXCESS AUTHORITY

Local admin. Debug access. Persistent exceptions.



Match the hardening to the failure

Cloud disconnected	→	Protect policy + detect stale data
Unsafe driver admitted	→	Driver blocking + HVCI + patching
Credentials exposed	→	Isolation + session / auth hardening
Telemetry lost	→	Protected config + off-host evidence

Harder to disable. Still possible to go around.

PROTECT TRUST

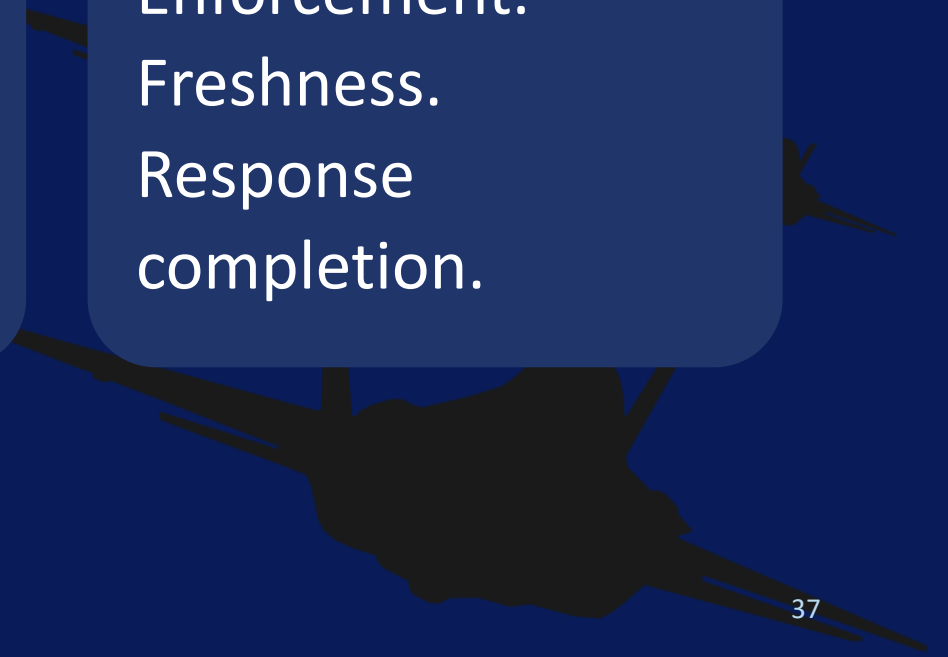
Boot. Kernel. Security-policy authority.

REDUCE OPTIONS

Privilege. Features. Exceptions.

PROVE COVERAGE

Enforcement. Freshness. Response completion.



Session materials

RESOURCES

GITHUB REPOSITORY

wpninja2026

github.com/schubarov/wpninja2026

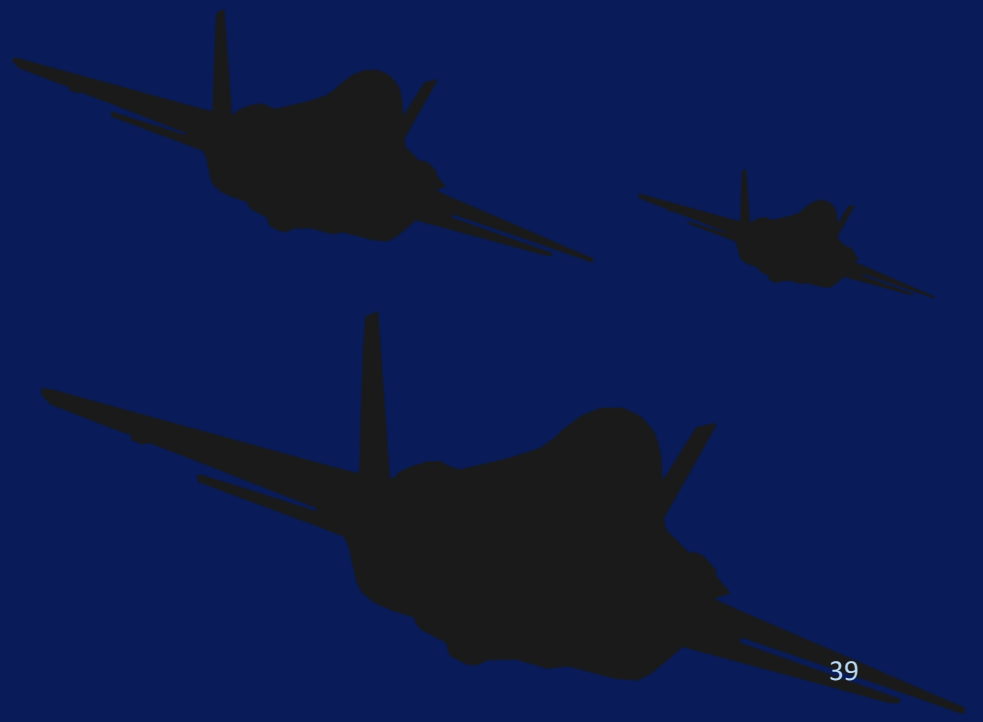
- Slides
- ASR exclusions + Lua bytecode
- Tools



SCAN TO OPEN ON GITHUB

Your feedback matters

Please rate this session.
Thank you for helping us improve.



Extended Q&A

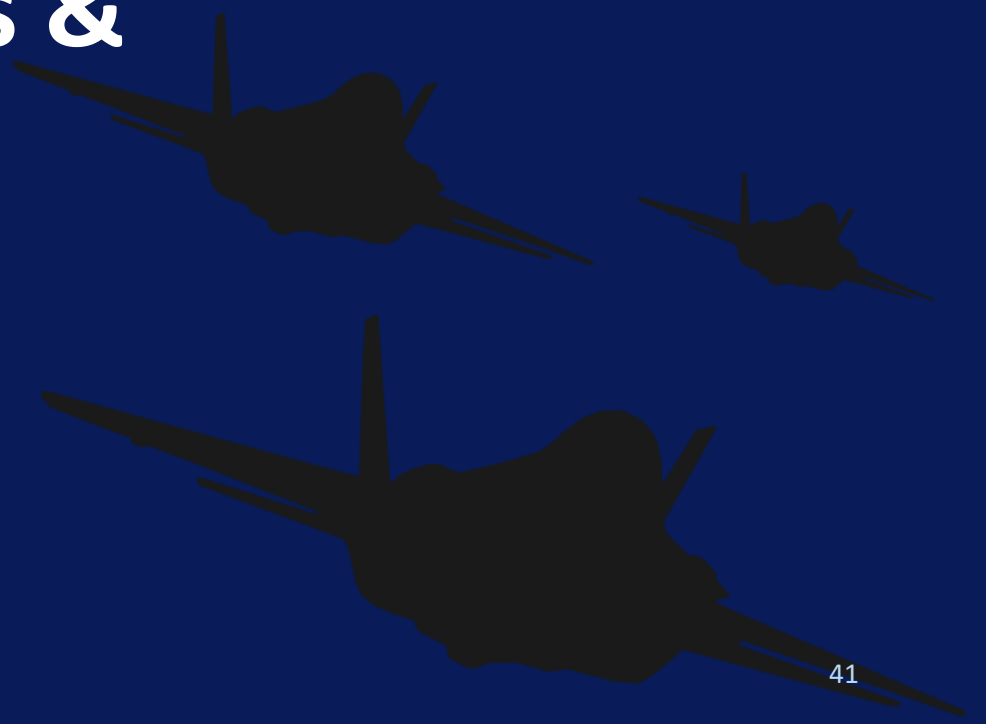
MMS



PATCH MY PC
robopack
ZeroTouch.ai
Recast
CODETWO
2Pint
POWERSTACKS
PDQ
ninjaOne
eido
App Control
flexera
THREATLOCKER
baramundi
nerdio
systemcenterdudes

Appendix: Additional Attacks & Mitigations

ADDITIONAL CASES & REFERENCES



A running process can still be an impaired sensor

FAILURE

Unresponsive processing.
Broken cloud delivery.

DEFENSE

Check real data freshness.
Verify completed actions.



Use the session—not the protected secret

OUTSIDE THE STORE

Application secrets. Existing identity authority.

REDUCE EXPOSURE

Separate admin sessions. Revoke exposed access.



VBS rollback: trusted does not mean current

HISTORICAL ISSUE

Older components could reintroduce fixed weaknesses.

MITIGATION

Current servicing. Applicable anti-rollback policy.



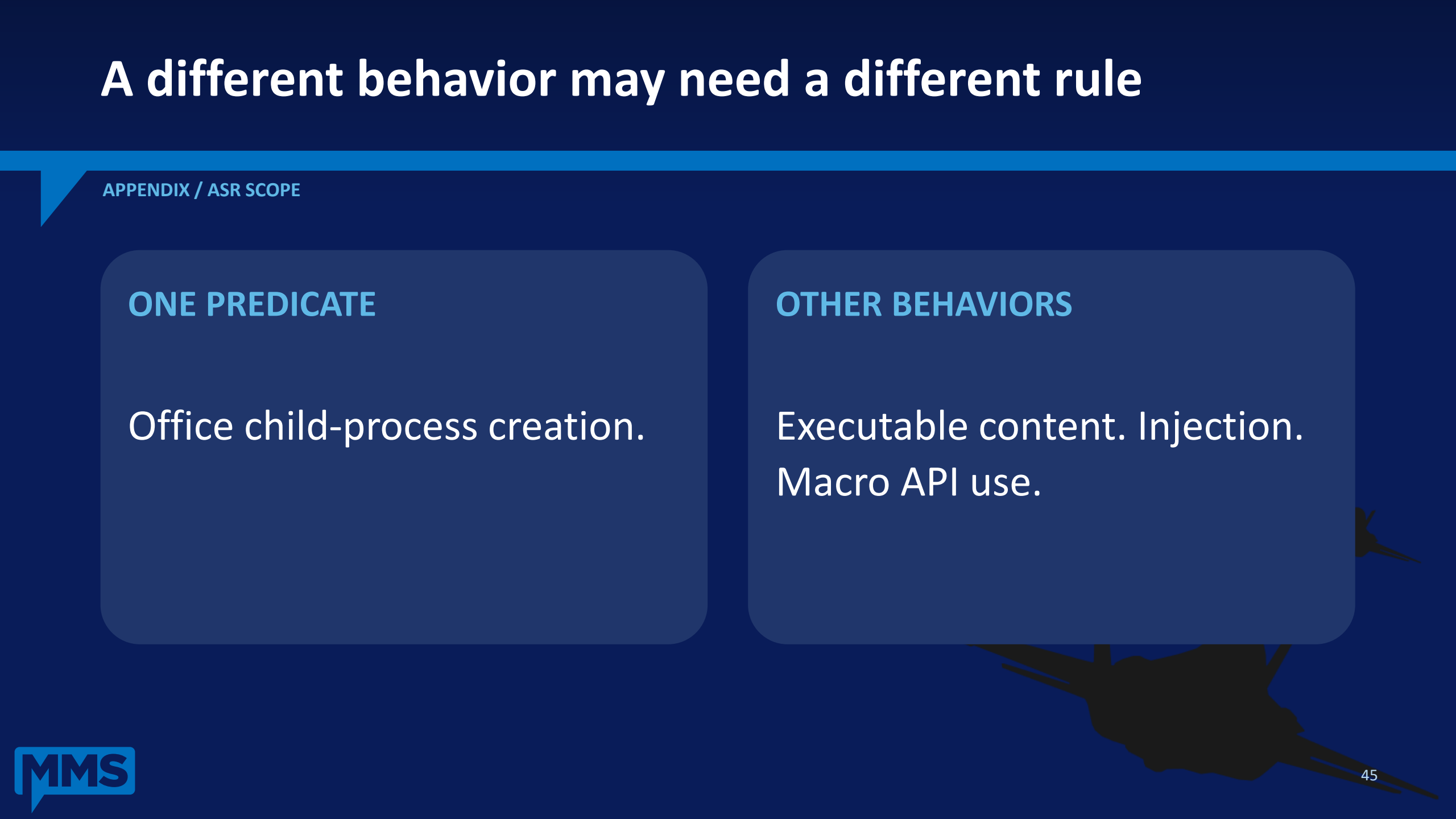
A different behavior may need a different rule

ONE PREDICATE

Office child-process creation.

OTHER BEHAVIORS

Executable content. Injection. Macro API use.



The event was never selected

COVERAGE GAP

Missing event type. Overbroad filter.

DEFENSE

Map detections to fields. Test benign markers.



Sources: EDR and Windows hardening

REFERENCES

R05 ESET - EDR killers and driverless impairment

R06 EDR architecture

R07 MDE connectivity

R01 Tamper protection

R09 Kernel-driver signing policy

R10 Attestation signing

R12 TechSpot - Terminator / Zemana reporting

R14 Adlice - RogueKiller product

R11 Vulnerable driver blocklist

R13 Memory Integrity / HVCI

R16 Secure Boot and Trusted Boot

Sources retrieved 09 Sep 2026. Full URLs are in speaker notes; titles are clickable.

Sources: credentials, ASR and Sysmon

REFERENCES

R03 Credential Guard architecture and protection limits

R02 Additional LSA protection

R18 Credential Guard state and configuration

R19 Additional identity mitigations

R26 KB5042562 / VBS anti-rollback protections

R04 ASR rule scope, dependencies and exceptions

R15 App Control overview

R22 Sysmon event and behavior reference

R23 Built-in Sysmon

R25 Sysmon configuration semantics

R24 Event forwarding and evidence collection

Sources retrieved 09 Sep 2026. Full URLs are in speaker notes; titles are clickable.